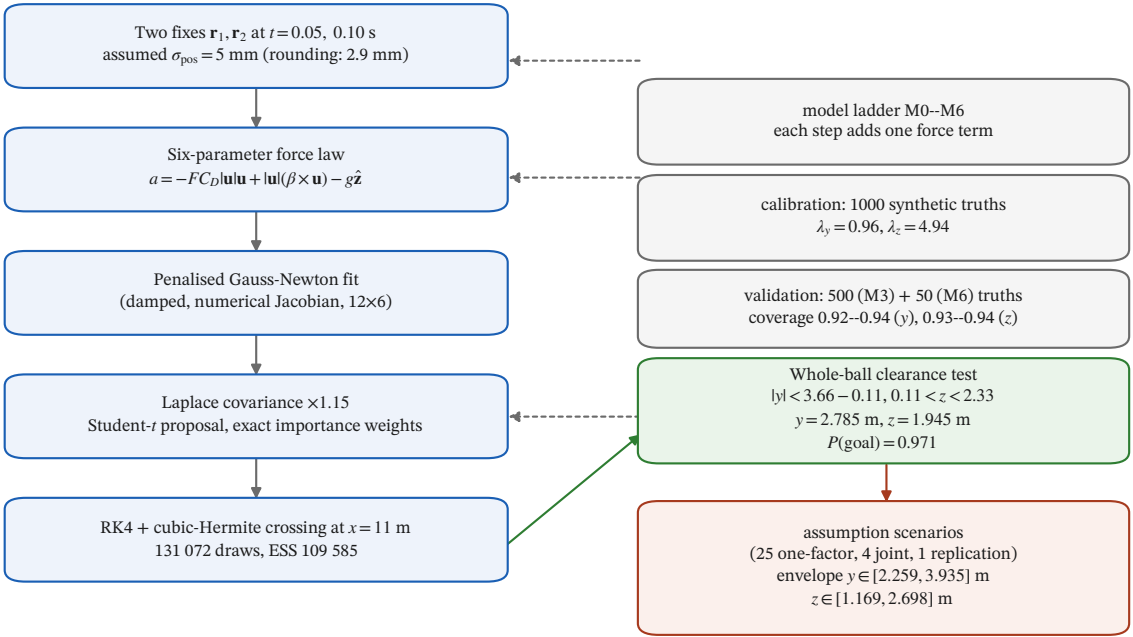
Two fixes $\mathbf{r}_1, \mathbf{r}_2$ at $t = 0.05,\ 0.10$ s
assumed $\sigma_{\text{pos}} = 5$ mm (rounding: 2.9 mm)
Six-parameter force law
$a = -FC_D|\mathbf{u}|\mathbf{u} + |\mathbf{u}|(\beta \times \mathbf{u}) - g\hat{\mathbf{z}}$
Penalised Gauss-Newton fit
(damped, numerical Jacobian, 12×6)
Laplace covariance ×1.15
Student-t proposal, exact importance weights
RK4 + cubic-Hermite crossing at $x = 11$ m
131 072 draws, ESS 109 585
model ladder M0--M6
each step adds one force term
calibration: 1000 synthetic truths
$\lambda_y = 0.96$, $\lambda_z = 4.94$
validation: 500 (M3) + 50 (M6) truths
coverage 0.92--0.94 (y), 0.93--0.94 (z)
Whole-ball clearance test
$|y| < 3.66 - 0.11$, $0.11 < z < 2.33$
$y = 2.785$ m, $z = 1.945$ m
$P(\text{goal}) = 0.971$
assumption scenarios
(25 one-factor, 4 joint, 1 replication)
envelope $y \in [2.259, 3.935]$ m
$z \in [1.169, 2.698]$ m